ECON UN3412 · Midterm cheat sheet

L1 · Introduction Sep 9

Causal = effect of *changing* X on Y. **Association** = X and Y move together (enough for prediction). Course question: is this association causal?

Data: **cross-section** (many units, once) · **panel** (same units, many times) · unstructured (text, images) must become a table first.

Randomized experiment = gold standard. Observational data: X not randomly assigned ⇒ need econometric tools.

Mozart effect: 36 college students, one spatial task, 15 minutes ⇒ says nothing about newborns, lasting IQ (external validity).

L2 · Probability I Sep 14

FORMULAS

$E(X) = \sum x_i \Pr(X = x_i)$ (long-run average; a constant; need not be a possible value)

$\mathbf{var}(X) = E[(X-\mu)^2] = E(X^2) - \mu^2$, $\sigma = \sqrt{\mathbf{var}}$ (same units as X)

Expectation	Variance
$E(c) = c$	$\mathbf{var}(c) = 0$
$E(aX+b) = aE(X) + b$	$\mathbf{var}(aX+b) = a^2\mathbf{var}(X)$
$E(X+Y) = EX + EY$ always	$\mathbf{var}(X+Y)$=sum of vars **if indep.**; else $+2\mathbf{cov}$

Dummy D: $E(D) = p$, $\mathbf{var}(D) = p(1-p)$. Die: E = 3.5, σ = 1.708. St. Petersburg: E = 1+1+⋯ = ∞.

Lawyers + AI (Autor 2026): randomized within firm; drafting +0.34–0.38 SD with AI; redlining without AI: seniors +0.45, juniors ≈ 0.

L3 · Probability II · estimators Sep 16

TWO VARIABLES

$\mathbf{cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)] = E(XY) - \mu_X\mu_Y$

$\mathbf{corr} = \mathbf{cov}/(\sigma_X\sigma_Y) \in [-1,1]$ (unit-free, **linear** only)

$\Pr(Y \mid X) = \dfrac{\Pr(X,Y)}{\Pr(X)}$

$E[Y \mid X = x] = \sum y \Pr(y \mid x)$ indep: $\Pr(X,Y) = \Pr(X)\Pr(Y)$

If $E[Y|X] = a + bX$: $b = \mathbf{cov}(X,Y)/\mathbf{var}(X)$ ← the regression slope. $\mathbf{cov}(aX+b, cY+d) = ac\,\mathbf{cov}$

0/1 SHORTCUT

$E(XY) = \Pr(X=1, Y=1)$ (XY = 1 only when both are 1). TikTok: cov = .15 − .55(.45) = −.0975, corr = −.39.

Estimand (fixed, unknown μ) · **estimator** (rule; random variable, e.g. $\bar{Y}$) · **estimate** (the number). Population = a data-generating process.

i.i.d.: independent (not roommate pairs) + identically distributed (not Columbia + NYU, not Dean's List only).

Unbiased: $E(\hat{\mu}) = \mu$ (right on average). **Precise**: small variance. **Consistent**: $\hat{\mu} \to \mu$ as n → ∞.

$E(\bar{Y}) = \mu$, $\mathbf{var}(\bar{Y}) = \sigma^2/n$ (independence kills cov terms). Y_1: unbiased, not consistent. $\frac{1}{n-1}\sum$: biased, consistent. $\bar{Y} + 0.1$: biased, not consistent. Weights summing to 1 ⇒ unbiased.

L4 · Large samples & inference Sep 21

- **LLN**: $\bar{Y} \to \mu$ (where it goes; also $s^2 \to \sigma^2$). **CLT**: $\bar{Y} \approx N(\mu, \sigma^2/n)$ for any population (shape); rule of thumb n > 30.
- LLN justifies sample means; CLT justifies CIs and tests even for non-normal data.

TEST IN SIX STEPS

1 $H_0 : \mu = \mu_0$, $H_1 : \mu \neq \mu_0$ (H_0 = default, burden on the claim) · 2 pretend H_0 true · 3 $t = \dfrac{\bar{Y} - \mu_0}{SE(\bar{Y})}$, $SE = s_Y/\sqrt{n}$ · 4 reject if $|t| > c$ · 5 $p = 2\Phi(-|t|)$ (one-sided $\Phi(-t)$) · 6 σ unknown ⇒ use s

α	10% *	5% **	1% ***
c	1.645	1.96	2.576
Φ(c)	.95	.975	.995

- **CI** = est ± c·SE = all μ_0 not rejected. 95% describes the *procedure*: 95% of such intervals contain μ.
- **p-value**: Pr(data this extreme *if H_0 true*). NOT Pr(H_0 true). Reject ⇔ p < α ⇔ |t| > c ⇔ μ_0 outside CI.
- **Type I** = reject true H_0 (prob α). **Type II** = miss a real effect. Lower α ⇒ more Type II. "Fail to reject," never "accept."
- Same t ⇒ same evidence (distance in SE units). 4× n ⇒ half the SE. Die: 4.2 avg, z = 2.24, p = .025.

L5 · Causality & potential outcomes Sep 23

DEFINITIONS

Y_{1i}, Y_{0i} = outcomes of the **same person** with/without treatment (not groups!). Effect $Y_{1i} - Y_{0i}$ never seen. $Y_i = Y_{0i} + (Y_{1i} - Y_{0i})D_i$. Target: $E[Y_{1i} - Y_{0i}]$.

KEY DECOMPOSITION

$$\underbrace{E[Y|D{=}1] - E[Y|D{=}0]}_{\text{observed difference}} = \underbrace{E[Y_1 - Y_0|D{=}1]}_{\text{effect on treated}} + \underbrace{E[Y_0|D{=}1] - E[Y_0|D{=}0]}_{\text{selection bias}}$$

SIGN THE BIAS

Would the treated do **better** without treatment? ⇒ + bias, overstated (gym, coffee, dogs, NHIS education). **Worse**? ⇒ − bias, understated or flipped (hospitals, Oregon insurance split).

- **Randomization** ⇒ D independent of Y_0, Y_1 ⇒ bias = 0 and effect on treated = ATE.
- $SE(\bar{Y}_1 - \bar{Y}_0) = \sqrt{s_1^2/n_1 + s_0^2/n_0}$. **Balance table**: checks pre-treatment traits; 1 in 20 flags by chance (10 tests ⇒ 40% chance of ≥1).
- **Oregon**: the treatment is *winning*, not enrolling. Medicaid +.256; good health +.039 (t = 4.9).
- **Internal** validity: right for those studied. **External**: carries to other people/places/times (Chile).
- "The design, not the size of the data set, makes a number causal."

L6 · Regression I: dummy = difference in means Sep 28

MODEL

$Y_i = \beta_0 + \beta_1 D_i + u_i$: $\hat{\beta}_0 = \bar{Y}_0$, $\hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0$. u = everything else (age, income, luck); $E[u] = 0$ is free (intercept absorbs it). $u_i = Y_{0i} - E[Y_{0i}]$.

- $E[u|D] = 0 \Leftrightarrow E[Y_0|D{=}1] = E[Y_0|D{=}0]$ ⇔ no selection bias (randomization gives it).
- **Bertrand–Mullainathan**: randomize names on résumés (can't randomize race). White 9.65% vs Black 6.45%, gap .032 (SE .0078), t = 4.1.

TAKE-UP AND SPILLOVERS

First stage = Pr(treated | offered) − Pr(treated | not offered). **Effect of treatment** = effect of offer ÷ first stage (needs **exclusion**: offer works only through treatment). Facebook: 10.1/.88 = 11.5 min; Oregon .039/.256 = .15.

Spillover: treat − control = effect − change in control. Control helped ⇒ too small; zero-sum (ranks) ⇒ too large. Internal-validity threat. Fix: randomize groups (clusters), costs precision.

- Facebook observational 0.23 SD vs experiment 0.09 ⇒ selection bias 0.14.

TRAPS TO AVOID (FROM YOUR NOTES)

- Reject when p < 0.05 (not >). |t| > 1.96 ⇔ reject.
- z and t **divide** by the SE: $(\bar{Y} - \mu_0)/SE$, using μ_0 not μ.
- Y_{1i}, Y_{0i} are the same person, not two groups; the outcome is Y, not D.
- $\Sigma\hat{u}_i = 0$ and $\Sigma X_i\hat{u}_i = 0$ are about **residuals** (always true), not errors.
- SER uses $(Y_i - \hat{Y}_i)^2$, not $(Y_i - \bar{Y})^2$. OLS (min SSR) maximizes R^2, not TSS.
- Consistency needs cov(X, **u**) = 0 and $E[u|X] = 0$, not $E[u] = 0$.
- LLN = where $\bar{Y}$ goes; CLT = its bell shape. SE ≠ SD of the data (SE = s/√n).
- 2× n ⇒ SE ÷ √2 (not ÷ 2). 4× n halves it.
- CI: 95% is about the method, not "95% chance μ is in this interval."
- Logs: earnings incl. zeros and profit (can be < 0) stay in levels.

Problem types · front (L2–L6) WS + PS

Type	Do this	Example
≥1 false flag in k tests	Pr = 1 − (1 − α)^k; expected # = kα	10 tests, 5%: 1 − .95^{10} = **.40**, expect 0.5. At 1%: 1 − .99^{10} = .10. 8 tests: .34; 9: .37
One flagged balance row	Not proof of failure: check joint test, size, procedure; add as control	Allcott joint p = .95; Oregon female p = .02, joint .24
Balance row t	diff ÷ √(s_1^2/n_1 + s_0^2/n_0); table shows SDs, not SEs	Age: .70/√(12.54²/580 + 11.71²/1081) = .70/.631 = 1.11, p = .27
Test a proportion	SE = √(p̂(1−p̂)/n), t = (p̂ − p_0)/SE	Poll 270/500: .54, SE .0223, t 1.79, p .073 (10% yes, 5% no). Cards: .338 vs .5, t −4.2
Pr($\bar{Y}$ in a range)	z = (a − μ)/(σ/√n), use Φ	Die n = 30: Pr($\bar{Y}$ > 4) = 1 − Φ(1.60) = .055
n for a target SE	n = (σ / SE)²	(1.708/.10)² ≈ 292
CI at 3 levels	est ± 1.645 / 1.96 / 2.576 × SE; 2× n ⇒ width ÷ √2	22.64 ± 1.96(1.28) = [20.13, 25.15]
Joint table	marginals = row/col sums; Pr(Y|X) = cell/marginal; 0/1: E = p, var = p(1−p), E(XY) = Pr(1,1)	Yerba: E(XY) = 1(.40) + 2(.10) = .60, cov = .60 − .95(.6) = .03
Linear transforms	E(aX+b) = aEX+b; cov(aY+b, cX+d) = ac·cov	W = 3Y − 2, V = 5 − 2X: cov = (3)(−2)(.03) = −.18
Nominal → real	× CPI_new/CPI_old (mean and SD both scale)	×237/156.9; real HS wage change CI includes 0
Potential-outcomes table	Y = Y_1 if D = 1 else Y_0; obs diff = ATT + bias	4 students: 2 = 1 + 1; lottery ⇒ ATE = 1 on average
Take-up	FS = share treated (offered − not); effect = ITT ÷ FS	Tutoring: FS .55, 3/.55 = 5.45; t = 3/1.2 = 2.5
Endowment logic	50% get A, 50% B ⇒ .5(1−X) + .5X = 50% trade	Actual 33.8% ⇒ endowment effect

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L7 · Regression II: least squares Sep 30

OLS

$\min \sum(Y_i - \hat\beta_0 - \hat\beta_1 X_i)^2 \Rightarrow$

$\hat\beta_1 = \dfrac{\sum(X_i-\bar X)(Y_i-\bar Y)}{\sum(X_i-\bar X)^2} = \dfrac{s_{XY}}{s_X^2}$, $\hat\beta_0 = \bar Y - \hat\beta_1 \bar X$

$\hat Y_i = \hat\beta_0 + \hat\beta_1 X_i$, $\hat u_i = Y_i - \hat Y_i$; $\sum \hat u_i = 0$, $\sum X_i \hat u_i = 0$; line passes through $(\bar X, \bar Y)$

Error u (to population line, never seen) ≠ **residual** û (to our line). "Residuals uncorrelated with X" is mechanical; "errors uncorrelated with X" is the assumption.

Slope: "units with one more [X] have β_1 more [Y], on average." College: $75.6/SAT pt = $7,562/100 pts.

Intercept: Ŷ at X = 0; meaningless if X = 0 is outside the data.

Why square? Sum of residuals can't rank lines; squares stop cancelling and punish big misses.

Hand example X = 10, 12, 14; Y = 45, 60, 66 ⇒ β_1 = 42/8 = 5.25, β_0 = −6.

L8 · Fit and logs Oct 7

FIT

TSS = ESS + SSR. $R^2 = \dfrac{ESS}{TSS} = 1 - \dfrac{SSR}{TSS} \in [0,1]$

(bivariate: corr²). $SER = \sqrt{\dfrac{SSR}{n-2}}$ = typical miss in Y units (Stata Root MSE; R Residual SE). n − 2: two parameters fitted.

FIT ≠ CAUSATION

Low R² can be causal (Oregon); high R² can be biased. R² = 0.51 ≠ "selectivity causes half of earnings."

Model	X ↑ by	Y changes by
Linear-log $Y = \beta_0 + \beta_1 \ln X$	1%	$0.01\beta_1$ units
Log-linear $\ln Y = \beta_0 + \beta_1 X$	1 unit	$100\beta_1$% (exact $e^{\beta_1} - 1$)
Log-log $\ln Y = \beta_0 + \beta_1 \ln X$	1%	β_1% (elasticity)

Logs: % changes, tame right tail, constant % ⇒ straight line. Levels if Y can be 0/negative (profit, all-adult earnings), a rate/score (GPA), or dollars are the question. ln(1+Y) fixes zeros only.

L9 · Properties of OLS Oct 12

KEY FORMULA

$\hat\beta_1 = \beta_1 + \dfrac{\sum(X_i-\bar X)u_i}{\sum(X_i-\bar X)^2} \xrightarrow{p} \beta_1 + \dfrac{\text{cov}(X,u)}{\text{var}(X)}$

$\hat\beta_1$ is a random variable (new sample, new slope).

Unbiased and **consistent** if $E[u|X] = 0$ (stronger than $E[u] = 0$, which is free). If $\text{cov}(X,u) > 0$, β_1 settles *above* β_1: **more data never fixes bias**.

CLT: $\hat\beta_1 \approx N.\ t = (\hat\beta_1 - \beta_{1,0})/SE$, CI = $\hat\beta_1 \pm 1.96SE$.

$SE(\hat\beta_1) \approx SER/(\sqrt{n}\, s_X)$: smaller with more n, less noise, more spread in X.

Heteroskedastic: spread of u varies with X ⇒ always **robust SEs** (`, robust`; `vcovHC(fit,"HC1")`). College: $2.38 → $2.94.

L10 · Omitted variable bias Oct 14

TWO ARROWS

Leaving out W biases β_1 only if **(1) W affects Y** AND **(2) W moves with X**. Either missing ⇒ no bias (W uncorrelated with X just adds noise).

$$\hat\beta_1^{short} \xrightarrow{p} \beta_1 + \beta_2 \cdot \frac{\text{cov}(X,W)}{\text{var}(X)}$$

In any sample: **short = long + β_2 × (slope of W on X)**.

Bias sign	cov(X,W) > 0	cov(X,W) < 0	cov = 0
$\beta_2 > 0$	↑ upward	↓ downward	none
$\beta_2 < 0$	↓ downward	↑ upward	none
$\beta_2 = 0$	none	none	none

WORKED EXAMPLE: RETURN TO COLLEGE

Y = log wages, X = years of school, W = ability. Short slope 0.10; β_2 = 0.05 per SD ability; ability per extra year = cov/var = 0.4 SD.

(1) $\beta_2 > 0$, cov > 0 ⇒ **upward**. (2) Bias = 0.05 × 0.4 = 0.02 ⇒ **long = short − bias** = 0.10 − 0.02 = **0.08**.

(3) Short: one more year ↔ ~10% higher wages, partly because schooled people have more ability. Long: *same ability*, one more year ↔ ~8%. Two of the ten points were ability.

OVER/UNDERSTATED = SIZE

Upward bias on a + effect overstates; on a − effect understates harm. Bias can flip the sign (office hours, hospitals, police). β_2 is from the **long** regression (holding X fixed), not the raw gap (height/sex: 708 → 900).

- College: 77.78 = 59.91 + 10,757 × .001662 (parents' income = 23%). Reperfusion: 190 = 120 + (−17.8)(−3.9). Public: −6,714 = −693 + 75 × (−80).
- Controls fix only what's included; lotteries fix everything.

L11 · Multiple regression Oct 19

- $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + u$. β_1 = effect of X_1 **holding the others constant** (compare within groups with same W). Controls needn't be random or causal.
- **Conditional mean independence**: holding controls fixed, X_1 is as good as randomly assigned. More plausible than no controls, still an assumption.
- SAT slope 77.8 → 59.9 → 59.9 (+public) → 53.0 (+adm. rate) → 49.8 while R² .524 → .551: R² says nothing about bias.

PERFECT MULTICOLLINEARITY

One regressor = exact linear function of others (intercept is a hidden "1"): years & months; male & female; work share & non-work share; all season dummies. Can't estimate (÷0; "change male holding female fixed"). Drop one ⇒ **reference category**; coefficients = differences from it. Not perfect: age & age², income & wealth (imperfect ⇒ big SEs).

- Seasons: fall ref 3,345; spring +20.89 = 3,366; every column predicts the same season means.
- R² never falls when adding X; $\bar R^2 = 1 - \frac{n-1}{n-k-1}\frac{SSR}{TSS}$ can (and can be < 0; R² can't). Don't pick controls by R².

L12 · Tests, interactions, polynomials Oct 21

- One coefficient: same t-test, add "holding others fixed." Brazil: educ R$405 [364, 446]; ms vs Goiás +256, p = .017 (5% yes, 1% no).

INTERACTIONS

$Y = \beta_0 + \beta_1 D_1 + \beta_2 D_2 + \beta_3 D_1 D_2$: gap from D_1 = β_1 (D_2 = 0), $\beta_1 + \beta_3$ (D_2 = 1). Old–young: 267 vs 267 + 2,393.

$Y = \beta_0 + \beta_1 X + \beta_2 D + \beta_3 XD$: D = 0 line ($\beta_0$, β_1); D = 1 line ($\beta_0+\beta_2$, $\beta_1+\beta_3$). β_2 shifts intercept (gap at X = 0!), β_3 = difference in slopes. Test same effect: t on β_3. Same line: F-test $\beta_2 = \beta_3 = 0$.

QUADRATIC

$Y = \beta_0 + \beta_1 X + \beta_2 X^2$: one more unit from a: $\hat\beta_1 + \hat\beta_2(2a+1)$ (or predict at both, subtract). $\beta_2 < 0$ ⇒ ∩. Linear vs quadratic: t on β_2. Pay: 25→26 +R$147; 45→46 ≈ 0; peak $-\beta_1/(2\beta_2)$ ≈ 45.5.

- "Linear model" = linear in **parameters**; X², ln X, X·D all fine for OLS.
- Joint tests ($\beta_1 = \beta_2 = 0$) ⇒ **F**, not two t's. One restriction: F = t².

Reading output Stata | R

Stata	R	Meaning
Coefficient	Estimate	$\hat\beta$ (Y units per X unit)
Robust std. err.	Std. Error (coeftest + vcovHC)	sampling spread of $\hat\beta$
t, P>|t|	t value, Pr(>|t|)	test of $\beta = 0$ **only**
_cons	(Intercept)	Ŷ when all X = 0
Root MSE	Residual std. error	SER

- Other H_0 ($\beta = 80$): compute t yourself or check the CI. Back out SE = $\hat\beta$/t. Rescale Y by c ⇒ $\hat\beta$, SE, SER × c; t, p, R² same.
- `summary()` SEs are not robust. R stars: *** .001 ** .01 * .05 (papers: 1/5/10%). 2e-16 ≈ 0. NA row ⇒ perfect multicollinearity.

SENTENCES TO USE

Slope: "[Units] with one more [X] have [$\hat\beta$] more [Y] on average (holding [W] fixed)." **Dummy:** "[group] has [$\hat\beta$] more [Y] than [reference]." **Log-lin:** "one more [X] ↔ ~[100$\hat\beta$]% more [Y]." **Sig:** "t = $\hat\beta$/SE > 1.96 ⇒ reject $\beta = 0$ at 5%." **CI:** "95% confident population β is in [a, b]." **p:** "if H_0 true, this extreme [p·100]% of the time." **R²:** "X accounts for [R²·100]% of the variation in Y." **SER:** "predictions typically miss by [SER]." **Causal?** "only if E[u|X] = 0; [W] is in u, affects Y, moves with X ⇒ biased [up/down]."

Problem types · back (L7–L12) WS + PS

Type	Do this	Example
OLS by hand	table of X−X̄, Y−Ȳ, product, (X−X̄)²; $\hat\beta_1$ = Σprod/Σsq	X 1,2,3; Y 2,4,9: 7/2 = 3.5, $\hat\beta_0$ = 5 − 7 = −2
Predict / residual	Ŷ = $\hat\beta_0 + \hat\beta_1$X; û = Y − Ŷ (+ = above line); at X̄, Ŷ = Ȳ	Columbia: 87,272; û = +15,219
Test β = value ≠ 0	t = ($\hat\beta$ − value)/SE, or is value in CI?	Height: (708 − 1000)/50.4 = −5.8 ⇒ reject
Dummy regression	_cons = ref mean; slope = gap; flip dummy ⇒ new ref, same group means, sign flips	private = 55,659 + 6,714·private, R² same
Switch reference state	mean = intercept + coef; every column gives the same means	MT: 3,279 + 288 = 4,845 − 1,278 = 3,567
OVB number check	short = long + $\hat\beta_2$ × (slope of W on X); with dummy X, slope = diff in mean W	Reperf: 120.3 + (−17.8)(−3.92) = 190.2
Share explained by W	(short − long)/short	SAT on public: 6,021/6,714 = 90%
Interaction gaps	fill a 2×2 table of $\hat\beta$'s; gap = β_1 (+β_3)	Old–young: 267 vs 2,659
Quadratic change	Ŷ(a+1) − Ŷ(a) = $\hat\beta_1 + \hat\beta_2(2a+1)$; peak $-\beta_1/2\beta_2$	25→26: +147; 45→46: ≈0
Perfect multicollinearity?	write one regressor as a + b·others (X_0 = 1)	months = 12·years; $X_2 = X_1$ + 100X_0; constant regressor
Design a study	randomize what you can; measure first stage; mind spillovers (cluster)	College lottery; Facebook $102 offer

EVERY "IS IT CAUSAL?" QUESTION

1 What is compared? 2 What else differs (what's in u)? 3 Does it move with X and affect Y? 4 Sign it: too big or too small?